TARS4 — wiring guide

The 112 TARS wires (33 in 26 AWG, the rest in 30 AWG), in build order. Each step is tested on the bench before the next. Steps 1–5 are done outside TARS; a tube is fitted in its columns before the far ends of its wires are finished (section Joints), so steps 6–7 are tested with the columns joined. The board plans (renders/tars4_board_*.svg) give hole by hole, this guide gives both ends of every wire.

WIRE END CODE (THE SAME ON THE BOARD PLANS)

- **Soldered**: on a board hole (plan shown) or on a module pad / lead. Tin the wire, solder, heat-shrink over the joint if it is exposed.

DF — **Dupont, female contact** crimped (SN-28B, 30 AWG folded in two; two wires in one contact: together, without folding), slid into housing Xn at the cavity shown (cavity 1 = pin 1, paint dot).

DM — **Dupont, male contact**: the module side of an inline pair (pigtails of X3, X7, X15).

XHF — **JST-XH female** crimped, on the power block side (X10, X13, X14) or the cell side (X12).

XHM — **JST-XH male** crimped, the other side of the pair.



Wire colour, thick = 26 AWG, strokes = black heat-shrink rings at both ends (1 = I2S / PIR, 2 = I2C, 3 = microSD).



Power pairs: red / black = battery and 5 V, white / blue = regulated 3V3. Rails sharing red / black carry a **coloured band** (10 mm of coloured heat-shrink, shrunk 2–3 cm from each end, the same on the + and on its GND): none = cell B+ / B−, **white** = TP4056 OUT+ / OUT−, **yellow** = VPATH, **orange** = VSYS, **purple** = VUSB. Slip the band on before soldering or crimping.

DIAGRAM BLOCKS (FRAME COLOUR)

P — **Hand-wired board**: P1, P2 (stripboards), P3 S3 carrier, P4 power path (pad perfboards); plans renders/ tars4_board_*.svg.

M — **Module or part** under its name: TP4056, TPS63020, INMP441, AM312, microSD reader, switch, cells, speaker...

X — **Dupont connector**, one side (female or male); cavities are numbered from pin 1.

X — **JST-XH connector**, one side (female or male).

In the diagrams, TARS is seen from the front (left leg on the left), one column per TARS column, the tubes between them. A block = a board, a module or one side of a connector; each line = one wire end. A wire started at one step and finished later leaves the block with an arrow "→ st. n".

1 Power block, on the bench

left leg: TP4056, power-path board, TPS63020 soldered together · 23 wires finished · housings to fill: X10 F, X12 M, X13 F, X14 F, X16 M

Before

1. Power-path board: parts, bridges and GND bus per renders/tars4_board_power.svg (Q1 / Q5 with wire-wrap wire).
2. TP4056: R_PROG replaced by 2.4 k, then tested alone: 3.7 V bench supply on B+ / B−, current < 10 µA. Polarities: C9 + on IN+, D3 band (cathode) on IN+ (reversed, it shorts the 5 V). TPS63020: EN tied to VIN, PS/SYNC to GND.
3. Crowded pads (joints ≥ 0.65 mm from the board end): splice 1 cm before the pad the two VUSB wires of IN+ (to X10 and to the power-path board), the two B+ (to X12 and X16, with the B+ sense wire to the power-path board) and the two B−: a single conductor per pad, next to C9 / D3.
4. Solder the wires below (4–5 cm loops between the three boards), then crimp the XH contacts and slide them into X10, X12, X13 and X14.
5. X12 and X16 on the block side (M): while unplugged, capped with an empty female XH housing. Their pins carry B+ / B− as soon as a cell is plugged in, until step 7.

Bench test

- ☐ 3.7 V bench supply on X12 (block side, M): OUT+ ≈ B+; current without USB noted (D1 leakage adds to the TP4056 one: a few tens of µA possible; power_budget.md).
- ☐ X10 cavities 3–4 bridged (switch simulated): VSYS ≈ B+; 3.3 V on X13 cavity 3; VBAT/2 ≈ B+ / 2 on X14 cavity 1.
- ☐ 5 V on X10 cavities 1–2: charge LED, USB_DET ≈ 3 V on X14 cavity 2, Q1 off.

Step 1 · Power block, on the bench: diagram

Wires finished at this step (23)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut
W1	red 26 AWG, purple band	VUSB	**P4** power path · hole F8 (wire 1 of the plan)	TP4056 · IN+ (VUSB)	—	**10** → 13
W2	red 26 AWG, white band	OUT+	**P4** power path · hole B4 (wire 2 of the plan)	TP4056 · OUT+	—	**11** → 14
W3	black 26 AWG, white band	GND	**P4** power path · hole C1 (wire 7 of the plan)	TP4056 · OUT−	—	**11** → 14
W4	brown 30 AWG	VBAT/2 → B+	**P4** power path · hole J5 (wire 10 of the plan)	TP4056 · B+ pad (R2 1 MΩ in line at the pad end) R2 1 MΩ soldered in line ~2 cm from the other end, in heat-shrink	—	**11** → 14
W5	red 26 AWG, orange band	VSYS	**P4** power path · hole C3 (wire 5 of the plan)	TP63020 · VIN	—	**12** → 15
W6	black 26 AWG, orange band	GND	**P4** power path · hole D1 (wire 8 of the plan)	TP63020 · GND	—	**12** → 15
W7	white 30 AWG	3V3	**P4** power path · hole H8 (wire 13 of the plan)	TP63020 · VOUT	—	**12** → 15
W8	red 26 AWG, yellow band	VPATH	**P4** power path · hole A8 (wire 3 of the plan)	XHF **X10** J1 + SW1 · case 3 VPATH	—	**11** → 14
W9	red 26 AWG, orange band	VSYS	**P4** power path · hole D3 (wire 4 of the plan)	XHF **X10** J1 + SW1 · case 4 VSYS	—	**10** → 13
W10	red 26 AWG, orange band	VSYS	**P4** power path · hole C3 (wire 6 of the plan)	XHF **X13** power block outputs · case 1 VSYS	—	**11** → 14
W11	black 26 AWG, orange band	GND	**P4** power path · hole E1 (wire 9 of the plan)	XHF **X13** power block outputs · case 2 GND amp	—	**11** → 14
W12	brown 30 AWG	VBAT/2	**P4** power path · hole J3 (wire 11 of the plan)	XHF **X14** measurements · case 1 VBAT/2	—	**11** → 14
W13	orange 30 AWG	USB_DET	**P4** power path · hole E2 (wire 12 of the plan)	XHF **X14** measurements · case 2 USB_DET	—	**9** → 12
W14	brown 30 AWG, 3 rings	SD_EN	**P4** power path · hole H5 (wire 15 of the plan)	XHF **X14** measurements · case 3 SD_EN	—	**11** → 14
W15	brown 30 AWG, 1 ring	VSYS → VSENSE_B	**P4** power path · hole D4 (wire 16 of the plan)	XHF **X14** measurements · case 4 VSENSE_B R23 1 MΩ soldered in line ~2 cm from the board end, in heat-shrink	—	**10** → 13
W16	red 26 AWG, purple band	VUSB	TP4056 · IN+	XHF **X10** J1 + SW1 · case 1 VUSB	—	**9** → 12
W17	black 26 AWG, purple band	GND	TP4056 · IN−	XHF **X10** J1 + SW1 · case 2 GND	—	**9** → 12
W18	red 26 AWG	B+	TP4056 · B+	XHM **X12** left cell · case 1 B+ cell junction: two wires soldered on the B+ pad	—	**10** → 13
W19	black 26 AWG	B−	TP4056 · B−	XHM **X12** left cell · case 2 B− cell junction: two wires soldered on the B− pad	—	**10** → 13
W20	red 26 AWG	B+	TP4056 · B+	XHM **X16** right cell · case 1 B+ to X16, parked above the power-path board	—	**13** → 16
W21	black 26 AWG	B−	TP4056 · B−	XHM **X16** right cell · case 2 B−	—	**13** → 16
W22	white 26 AWG	3V3	TPS63020 · VOUT	XHF **X13** power block outputs · case 3 3V3 along the logic GND wire	—	**11** → 14
W23	blue 26 AWG	GND	TPS63020 · GND	XHF **X13** power block outputs · case 4 GND log logic GND, along the 3V3	—	**11** → 14

Wires started here, finished later (1)

One end only now; the other stays bare until the step shown.

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W59	**1→5**	white 30 AWG, 3 rings	SD_3V3	**P4** power path · hole I6 (wire 14 of the plan)	DF **X15** microSD · case 4 3V3	—	**16** → 19

2 Charge port, switch, cells

the foot and cell cables up to X10, X12 and X16 · 6 wires finished · housings to fill: X10 M, X12 F

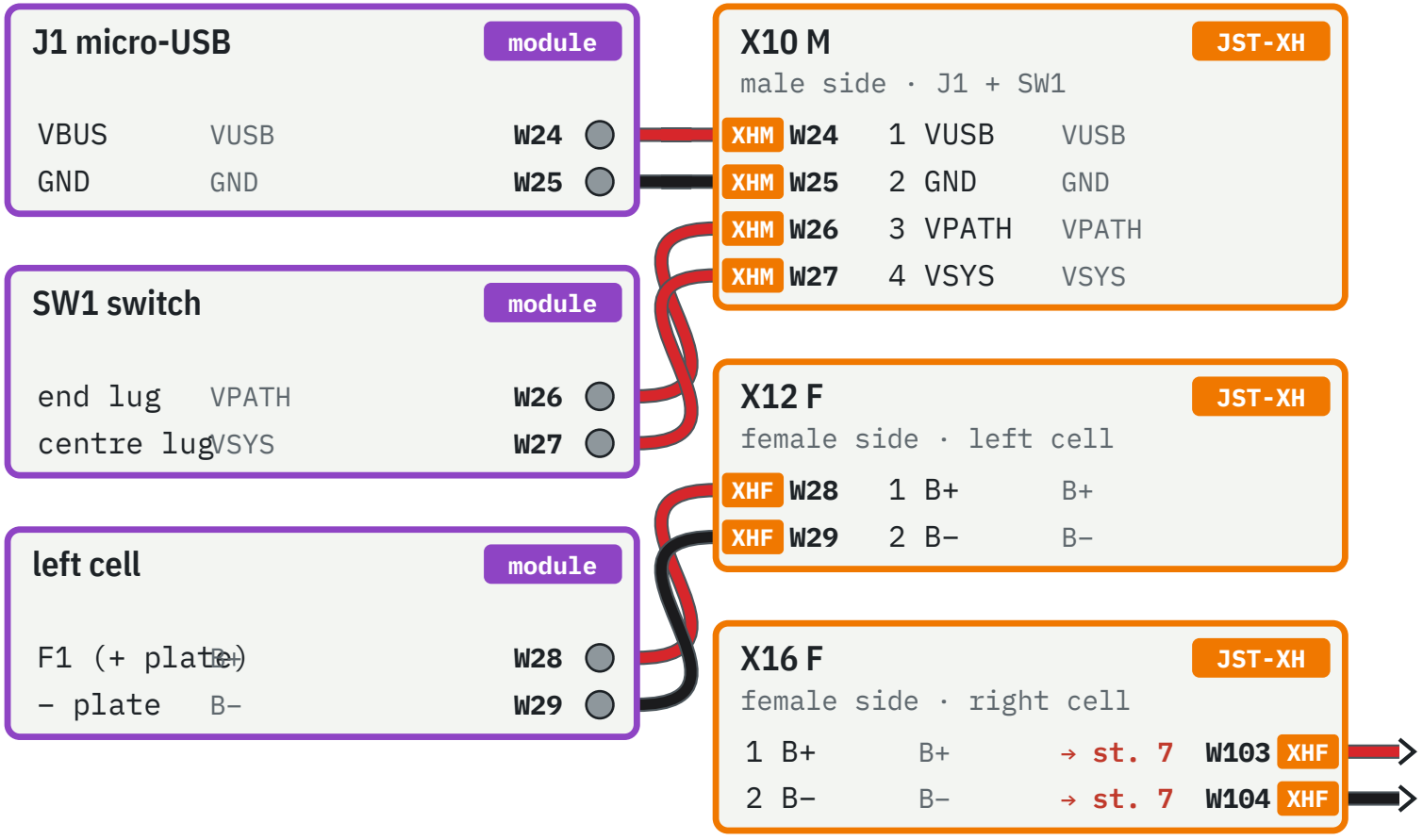
Before

1. Solder F1 on the left + plate (F2 on the right one at step 7). Never solder on a cell.
2. Right cell wires: long enough to cross the three tubes (length from the table, already + 3 cm, offered up on TARS before cutting). Crimp X16 (female) now; the right cell end stays bare (soldered at step 7, once the right tube is fitted).

Bench test

- ☐ Left cell alone: X12 plugged in, switch on, VSYS ≈ B+.
- ☐ Charger on J1: charge LED; unplugged: VUSB < 0.5 V within ~1 s.

LEFT LEG


J1 micro-USB
module
VBUS VUSB W24
GND GND W25
SW1 switch
module
end lug VPATH W26
centre lug VSYS W27
left cell
module
F1 (+ plate) B+ W28
− plate B− W29
X10 M
JST-XH
male side · J1 + SW1
XHM W24 1 VUSB VUSB
XHM W25 2 GND GND
XHM W26 3 VPATH VPATH
XHM W27 4 VSYS VSYS
X12 F
JST-XH
female side · left cell
XHF W28 1 B+ B+
XHF W29 2 B− B−
X16 F
JST-XH
female side · right cell
1 B+ B+ → st. 7 W103 XHF
2 B− B− → st. 7 W104 XHF

Wires finished at this step (6)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut
W24	 red 26 AWG, purple band	VUSB	● J1 micro-USB · VBUS	XHM **X10** J1 + SW1 · case 1 VUSB twisted with the GND	—	**24** → 27
W25	 black 26 AWG, purple band	GND	● J1 micro-USB · GND	XHM **X10** J1 + SW1 · case 2 GND	—	**24** → 27
W26	 red 26 AWG, yellow band	VPATH	● SW1 switch · end lug	XHM **X10** J1 + SW1 · case 3 VPATH	—	**24** → 27
W27	 red 26 AWG, orange band	VSYS	● SW1 switch · centre lug	XHM **X10** J1 + SW1 · case 4 VSYS	—	**24** → 27
W28	 red 26 AWG	B+	● left cell · F1 (+ plate)	XHF **X12** left cell · case 1 B+ twisted with B−	—	**18** → 21
W29	 black 26 AWG	B−	● left cell · − plate	XHF **X12** left cell · case 2 B−	—	**24** → 27

Wires started here, finished later (2)

One end only now; the other stays bare until the step shown.

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W103	2→7	 red 26 AWG	B+	● right cell · F2 (+ plate)	XHF **X16** right cell · case 1 B+ X16 crimped at step 2, bare end threaded through the 3 tubes, soldered on F2 at step 7; twisted with B−	left tube + middle tube + right tube	**52** → 55
W104	2→7	 black 26 AWG	B−	● right cell · − plate	XHF **X16** right cell · case 2 B− as B+; twisted with B+	left tube + middle tube + right tube	**57** → 60

3 S3 carrier

the 27 controller wires, soldered on the carrier side · 0 wires finished · housings to fill: none

Before

1. Plan renders/tars4_board_s3.svg: 2 × 8 headers (V contact removed, headers glued), standing parts, bridges, GND bus, jumpers.
2. Solder the 27 wires on the carrier side only, long enough (6–8 cm service loop + route). The other end stays bare: it is finished at the step shown.

Bench test

- ☐ Before any power-up: every still-free end capped with closed heat-shrink (shrunk, end pinched while hot).
- ☐ S3 removed: each wire ↔ its pin with the ohmmeter, GND isolated from 3V3 and the signals.
- ☐ S3 plugged in, PC USB: test program flashed, the S3 boots.

Step 3 · S3 carrier: diagram

Wires finished at this step (0)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut

Wires started here, finished later (28)

One end only now; the other stays bare until the step shown.

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W30	3→4	white 26 AWG	3V3	**P3** S3 carrier · hole A4 (wire 1 of the plan)	**P1** left middle distribution · hole B9 (wire 8 of the plan)	—	**28** → 31
W31	3→4	blue 26 AWG	GND	**P3** S3 carrier · hole J5 (wire 2 of the plan)	**P1** left middle distribution · hole B10 (wire 2 of the plan)	—	**30** → 33
W32	3→4	red 30 AWG	DC	**P3** S3 carrier · hole A7 (wire 5 of the plan)	**P1** left middle distribution · hole A7 (wire 15 of the plan)	—	**26** → 29
W33	3→4	green 30 AWG	MOSI	**P3** S3 carrier · hole C10 (wire 6 of the plan)	**P1** left middle distribution · hole A5 (wire 21 of the plan)	—	**25** → 28
W34	3→4	brown 30 AWG	BL_EN	**P3** S3 carrier · hole B7 (wire 12 of the plan)	**P1** left middle distribution · hole A8 (wire 12 of the plan)	—	**27** → 30
W35	3→4	orange 30 AWG	RST	**P3** S3 carrier · hole B6 (wire 13 of the plan)	**P1** left middle distribution · hole A6 (wire 18 of the plan)	—	**26** → 29
W36	3→4	purple 30 AWG	CS1	**P3** S3 carrier · hole A5 (wire 7 of the plan)	DF **X1** screen 1 · case 7 CS	—	**36** → 39
W37	3→4	yellow 30 AWG	SCK1	**P3** S3 carrier · hole C7 (wire 16 of the plan)	DF **X1** screen 1 · case 3 SCL	—	**35** → 38
W38	3→4	red 30 AWG	INT1	**P3** S3 carrier · hole A10 (wire 3 of the plan)	DF **X2** ADXL345 · case 4 INT1	—	**22** → 25
W39	3→4	green 30 AWG, 2 rings	SDA	**P3** S3 carrier · hole B9 (wire 10 of the plan)	DF **X2** ADXL345 · case 7 SDA	—	**22** → 25
W40	3→4	yellow 30 AWG, 2 rings	SCL	**P3** S3 carrier · hole B8 (wire 11 of the plan)	DF **X2** ADXL345 · case 8 SCL	—	**22** → 25
W41	3→4	yellow 30 AWG, 1 ring	MIC_BCLK	**P3** S3 carrier · hole B4 (wire 15 of the plan)	DF **X3** INMP441 · case 1 SCK	—	**24** → 27
W42	3→4	purple 30 AWG, 1 ring	MIC_WS	**P3** S3 carrier · hole I9 (wire 18 of the plan)	DF **X3** INMP441 · case 2 WS	—	**22** → 25
W43	3→4	green 30 AWG, 1 ring	MIC_SD	**P3** S3 carrier · hole I4 (wire 21 of the plan)	DF **X3** INMP441 · case 3 SD	—	**23** → 26
W60	3→5	brown 30 AWG	VBAT/2	**P3** S3 carrier · hole B11 (wire 8 of the plan)	XHM **X14** measurements · case 1 VBAT/2	left tube	**50** → 53
W61	3→5	orange 30 AWG	USB_DET	**P3** S3 carrier · hole I7 (wire 20 of the plan)	XHM **X14** measurements · case 2 USB_DET	left tube	**53** → 56
W62	3→5	brown 30 AWG, 3 rings	SD_EN	**P3** S3 carrier · hole J9 (wire 22 of the plan)	XHM **X14** measurements · case 3 SD_EN	left tube	**52** → 55
W63	3→5	brown 30 AWG, 1 ring	VSENSE_B	**P3** S3 carrier · hole G8 (wire 28 of the plan)	XHM **X14** measurements · case 4 VSENSE_B	left tube	**52** → 55
W64	3→5	purple 30 AWG, 3 rings	SD_CMD	**P3** S3 carrier · hole I8 (wire 19 of the plan)	DF **X15** microSD · case 2 CMD	left tube	**57** → 60
W65	3→5	green 30 AWG, 3 rings	SD_D0	**P3** S3 carrier · hole J8 (wire 23 of the plan)	DF **X15** microSD · case 3 D0	left tube	**57** → 60
W66	3→5	yellow 30 AWG, 3 rings	SD_CLK	**P3** S3 carrier · hole G11 (wire 25 of the plan)	DF **X15** microSD · case 1 CLK	left tube	**55** → 58
W75	3→6	brown 30 AWG, 1 ring	AMP_SD_S3	**P3** S3 carrier · hole A9 (wire 4 of the plan)	**P2** right middle distribution · hole A11 (wire 1 of the plan)	middle tube	**49** → 52
W76	3→6	red 30 AWG, 1 ring	PIR	**P3** S3 carrier · hole B10 (wire 9 of the plan)	**P2** right middle distribution · hole A4 (wire 15 of the plan)	middle tube	**51** → 54
W77	3→6	purple 30 AWG	CS2	**P3** S3 carrier · hole B5 (wire 14 of the plan)	DF **X4** screen 2 · case 7 CS	middle tube	**47** → 50
W78	3→6	yellow 30 AWG	SCK2	**P3** S3 carrier · hole D7 (wire 17 of the plan)	DF **X4** screen 2 · case 3 SCL	middle tube	**46** → 49

W79	**3→6**	 green 30 AWG, 1 ring	DIN	**P3** S3 carrier · hole J6 (wire 24 of the plan)	**DF**	**X5** MAX98357A · case 3 DIN	middle tube	**57** → 60
W80	**3→6**	 purple 30 AWG, 1 ring	LRC	**P3** S3 carrier · hole G6 (wire 26 of the plan)	**DF**	**X5** MAX98357A · case 1 LRC	middle tube	**58** → 61
W81	**3→6**	 yellow 30 AWG, 1 ring	BCLK	**P3** S3 carrier · hole G4 (wire 27 of the plan)	**DF**	**X5** MAX98357A · case 2 BCLK	middle tube	**59** → 62

4 P1 and the left middle

screen 1, mic, accelerometer · 29 wires finished · housings to fill: X1 F, X2 F, X3 F, X3 M

Before

1. Plan renders/tars4_board_p1.svg. 1 × 8 right-angle header on the back of screen 1 (X1). ADXL345: straight header, INT2 removed and its cavity plugged (X2).
2. INMP441: 5-wire pigtail (male contacts, X3); L/R bridged to GND on its board.
3. X2 SDA and SCL: double contact, crimp together (without folding back) the S3 wire and the one leaving for P2 (bare end, soldered at step 6).

Bench test

- ☐ Before any power-up: every still-free end capped with closed heat-shrink (shrunk, end pinched while hot).
- ☐ S3 on USB, switch off: screen 1 displays, the mic gives I2S samples, the ADXL345 answers at 0x53 and INT1 toggles.

Step 4 · P1 and the left middle: diagram

Wires finished at this step (29)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut
W30	white 26 AWG	3V3	**P3** S3 carrier · hole A4 (wire 1 of the plan)	**P1** left middle distribution · hole B9 (wire 8 of the plan)	—	**28** → 31
W31	blue 26 AWG	GND	**P3** S3 carrier · hole J5 (wire 2 of the plan)	**P1** left middle distribution · hole B10 (wire 2 of the plan)	—	**30** → 33
W32	red 30 AWG	DC	**P3** S3 carrier · hole A7 (wire 5 of the plan)	**P1** left middle distribution · hole A7 (wire 15 of the plan)	—	**26** → 29
W33	green 30 AWG	MOSI	**P3** S3 carrier · hole C10 (wire 6 of the plan)	**P1** left middle distribution · hole A5 (wire 21 of the plan)	—	**25** → 28
W34	brown 30 AWG	BL_EN	**P3** S3 carrier · hole B7 (wire 12 of the plan)	**P1** left middle distribution · hole A8 (wire 12 of the plan)	—	**27** → 30
W35	orange 30 AWG	RST	**P3** S3 carrier · hole B6 (wire 13 of the plan)	**P1** left middle distribution · hole A6 (wire 18 of the plan)	—	**26** → 29
W36	purple 30 AWG	CS1	**P3** S3 carrier · hole A5 (wire 7 of the plan)	DF **X1** screen 1 · case 7 CS	—	**36** → 39
W37	yellow 30 AWG	SCK1	**P3** S3 carrier · hole C7 (wire 16 of the plan)	DF **X1** screen 1 · case 3 SCL	—	**35** → 38
W38	red 30 AWG	INT1	**P3** S3 carrier · hole A10 (wire 3 of the plan)	DF **X2** ADXL345 · case 4 INT1	—	**22** → 25
W39	green 30 AWG, 2 rings	SDA	**P3** S3 carrier · hole B9 (wire 10 of the plan)	DF **X2** ADXL345 · case 7 SDA	—	**22** → 25
W40	yellow 30 AWG, 2 rings	SCL	**P3** S3 carrier · hole B8 (wire 11 of the plan)	DF **X2** ADXL345 · case 8 SCL	—	**22** → 25
W41	yellow 30 AWG, 1 ring	MIC_BCLK	**P3** S3 carrier · hole B4 (wire 15 of the plan)	DF **X3** INMP441 · case 1 SCK	—	**24** → 27
W42	purple 30 AWG, 1 ring	MIC_WS	**P3** S3 carrier · hole I9 (wire 18 of the plan)	DF **X3** INMP441 · case 2 WS	—	**22** → 25
W43	green 30 AWG, 1 ring	MIC_SD	**P3** S3 carrier · hole I4 (wire 21 of the plan)	DF **X3** INMP441 · case 3 SD	—	**23** → 26
W44	yellow 30 AWG, 1 ring	MIC_BCLK	INMP441 · SCK	DM **X3** INMP441 · case 1 SCK 5–6 cm pigtail	—	**6** → 7
W45	purple 30 AWG, 1 ring	MIC_WS	INMP441 · WS	DM **X3** INMP441 · case 2 WS	—	**6** → 7
W46	green 30 AWG, 1 ring	MIC_SD	INMP441 · SD	DM **X3** INMP441 · case 3 SD	—	**6** → 7
W47	white 30 AWG	3V3	INMP441 · VDD	DM **X3** INMP441 · case 4 VDD	—	**6** → 7
W48	blue 30 AWG	GND	INMP441 · GND	DM **X3** INMP441 · case 5 GND L/R bridged to GND on the board	—	**6** → 7
W49	blue 30 AWG	GND	**P1** left middle distribution · hole A10 (wire 1 of the plan)	DF **X1** screen 1 · case 1 GND	—	**15** → 18
W50	white 30 AWG	3V3	**P1** left middle distribution · hole A9 (wire 7 of the plan)	DF **X1** screen 1 · case 2 VCC	—	**15** → 18
W51	brown 30 AWG	BL_EN	**P1** left middle distribution · hole B8 (wire 13 of the plan)	DF **X1** screen 1 · case 8 BLK	—	**15** → 18
W52	red 30 AWG	DC	**P1** left middle distribution · hole B7 (wire 16 of the plan)	DF **X1** screen 1 · case 6 DC	—	**15** → 18
W53	orange 30 AWG	RST	**P1** left middle distribution · hole B6 (wire 19 of the plan)	DF **X1** screen 1 · case 5 RES	—	**15** → 18
W54	green 30 AWG	MOSI	**P1** left middle distribution · hole B5 (wire 22 of the plan)	DF **X1** screen 1 · case 4 SDA	—	**16** → 19
W55	blue 30 AWG	GND	**P1** left middle distribution · hole D11 (wire 6 of the plan)	DF **X3** INMP441 · case 5 GND	—	**16** → 19
W56	white 30 AWG	3V3	**P1** left middle distribution · hole D9 (wire 10 of the plan)	DF **X3** INMP441 · case 4 VDD	—	**15** → 18

W57	white 30 AWG	3V3	DF **X3** INMP441 · case 4 VDD	DF **X2** ADXL345 · case 2 VCC double contact in X3 (with the wire from P1)	–	**12** → 15
W58	blue 30 AWG	GND	DF **X3** INMP441 · case 5 GND	DF **X2** ADXL345 · case 1 GND double contact in X3 (with the wire from P1)	–	**12** → 15

Wires started here, finished later (11)

One end only now; the other stays bare until the step shown.

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W67	4→5	blue 26 AWG	GND	● **P1** left middle distribution · hole C10 (wire 3 of the plan)	XHM **X13** power block outputs · case 4 GND log	left tube	**33** → 36
W68	4→5	white 26 AWG	3V3	● **P1** left middle distribution · hole C9 (wire 9 of the plan)	XHM **X13** power block outputs · case 3 3V3	left tube	**33** → 36
W69	4→5	blue 30 AWG, 3 rings	GND	● **P1** left middle distribution · hole B11 (wire 4 of the plan)	DF **X15** microSD · case 5 GND	left tube	**35** → 38
W82	4→6	blue 30 AWG	GND	● **P1** left middle distribution · hole C11 (wire 5 of the plan)	● **P2** right middle distribution · hole A10 (wire 3 of the plan)	middle tube	**29** → 32
W83	4→6	white 30 AWG	3V3	● **P1** left middle distribution · hole F9 (wire 11 of the plan)	● **P2** right middle distribution · hole A9 (wire 7 of the plan)	middle tube	**29** → 32
W84	4→6	brown 30 AWG	BL_EN	● **P1** left middle distribution · hole C8 (wire 14 of the plan)	DF **X4** screen 2 · case 8 BLK	middle tube	**27** → 30
W85	4→6	red 30 AWG	DC	● **P1** left middle distribution · hole C7 (wire 17 of the plan)	DF **X4** screen 2 · case 6 DC	middle tube	**27** → 30
W86	4→6	orange 30 AWG	RST	● **P1** left middle distribution · hole C6 (wire 20 of the plan)	DF **X4** screen 2 · case 5 RES	middle tube	**27** → 30
W87	4→6	green 30 AWG	MOSI	● **P1** left middle distribution · hole C5 (wire 23 of the plan)	DF **X4** screen 2 · case 4 SDA	middle tube	**28** → 31
W88	4→6	green 30 AWG, 2 rings	SDA	● **P2** right middle distribution · hole A8 (wire 10 of the plan)	DF **X2** ADXL345 · case 7 SDA	middle tube	**36** → 39
W89	4→6	yellow 30 AWG, 2 rings	SCL	● **P2** right middle distribution · hole A7 (wire 12 of the plan)	DF **X2** ADXL345 · case 8 SCL	middle tube	**37** → 40

5 Left tube: power block ↔ left middle, microSD

everything crossing the left tube · 16 wires finished · housings to fill: X13 M, X14 M, X15 F, X15 M

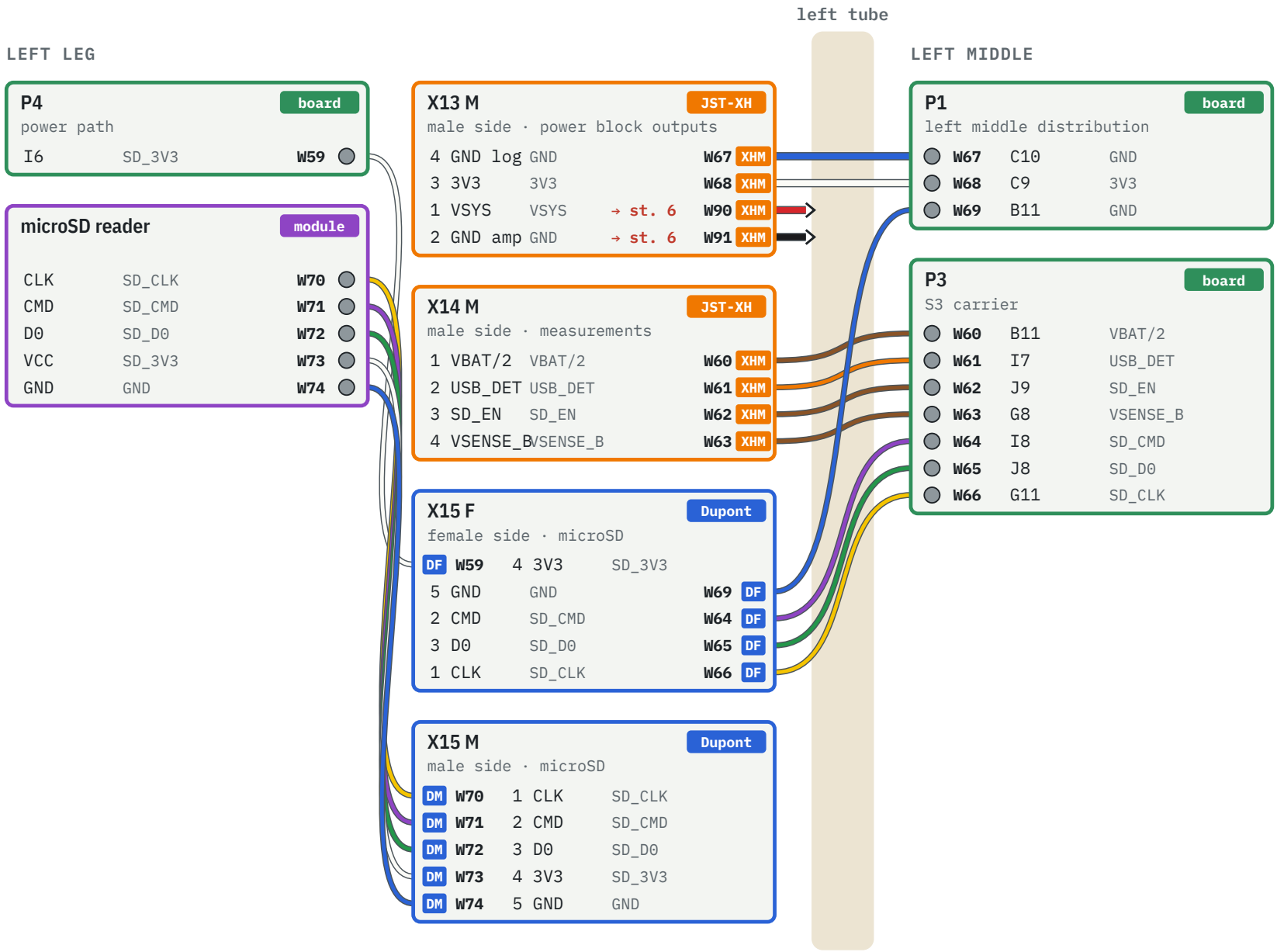
Before

1. Thread the left brass tube onto the listed wires, bare, BEFORE crimping their left leg end. The amplifier pair (step 6) and the right cell pair (step 2) also cross this tube: thread them at the same time.
2. After the bench test, the tube-side contacts of X13, X14, X15 and X16 come out of their housings again (needle under the latch) to fit the left tube (section Joints), then go back in.
3. microSD reader: C4 10 µF tantalum lying flat (+ on VCC; only ~6.6 mm up to the TPS63020) and C5 100 nF on its VCC / GND pins; without pull-ups on the reader, R5–R9 10 k from CMD, D0, D1, D2 and D3 to VCC (SD_3V3). Then a 5-wire pigtail (male contacts, X15).

Bench test

- ☐ Before any power-up: every still-free end capped with closed heat-shrink (shrunk, end pinched while hot).
- ☐ Without USB, switch on: the S3 boots from the cells (3.3 V on the carrier).
- ☐ VBAT/2 and USB_DET read by the S3; the microSD is read; SD_EN high cuts its 3.3 V.

Step 5 · Left tube: power block ⇔ left middle, microSD: diagram



Wires finished at this step (16)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut
W59	white 30 AWG, 3 rings	SD_3V3	**P4** power path · hole I6 (wire 14 of the plan)	DF **X15** microSD · case 4 3V3	—	**16** → 19
W60	brown 30 AWG	VBAT/2	**P3** S3 carrier · hole B11 (wire 8 of the plan)	XHM **X14** measurements · case 1 VBAT/2	left tube	**50** → 53
W61	orange 30 AWG	USB_DET	**P3** S3 carrier · hole I7 (wire 20 of the plan)	XHM **X14** measurements · case 2 USB_DET	left tube	**53** → 56
W62	brown 30 AWG, 3 rings	SD_EN	**P3** S3 carrier · hole J9 (wire 22 of the plan)	XHM **X14** measurements · case 3 SD_EN	left tube	**52** → 55
W63	brown 30 AWG, 1 ring	VSENSE_B	**P3** S3 carrier · hole G8 (wire 28 of the plan)	XHM **X14** measurements · case 4 VSENSE_B	left tube	**52** → 55
W64	purple 30 AWG, 3 rings	SD_CMD	**P3** S3 carrier · hole I8 (wire 19 of the plan)	DF **X15** microSD · case 2 CMD	left tube	**57** → 60
W65	green 30 AWG, 3 rings	SD_D0	**P3** S3 carrier · hole J8 (wire 23 of the plan)	DF **X15** microSD · case 3 D0	left tube	**57** → 60
W66	yellow 30 AWG, 3 rings	SD_CLK	**P3** S3 carrier · hole G11 (wire 25 of the plan)	DF **X15** microSD · case 1 CLK	left tube	**55** → 58
W67	blue 26 AWG	GND	**P1** left middle distribution · hole C10 (wire 3 of the plan)	XHM **X13** power block outputs · case 4 GND log	left tube	**33** → 36
W68	white 26 AWG	3V3	**P1** left middle distribution · hole C9 (wire 9 of the plan)	XHM **X13** power block outputs · case 3 3V3	left tube	**33** → 36
W69	blue 30 AWG, 3 rings	GND	**P1** left middle distribution · hole B11 (wire 4 of the plan)	DF **X15** microSD · case 5 GND	left tube	**35** → 38
W70	yellow 30 AWG, 3 rings	SD_CLK	microSD reader · CLK	DM **X15** microSD · case 1 CLK 5–6 cm pigtail	—	**6** → 7
W71	purple 30 AWG, 3 rings	SD_CMD	microSD reader · CMD	DM **X15** microSD · case 2 CMD	—	**6** → 7
W72	green 30 AWG, 3 rings	SD_D0	microSD reader · D0	DM **X15** microSD · case 3 D0	—	**6** → 7
W73	white 30 AWG, 3 rings	SD_3V3	microSD reader · VCC	DM **X15** microSD · case 4 3V3	—	**6** → 7
W74	blue 30 AWG, 3 rings	GND	microSD reader · GND	DM **X15** microSD · case 5 GND	—	**6** → 7

Wires started here, finished later (2)

One end only now; the other stays bare until the step shown.

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W90	5→6	red 26 AWG, orange band	VSYS	XHM **X13** power block outputs · case 1 VSYS	DF **X5** MAX98357A · case 7 VIN amplifier pair: threaded through the left tube at step 5, X5 end crimped here; twisted with its GND	left tube + middle tube	**48** → 51
W91	5→6	black 26 AWG, orange band	GND	XHM **X13** power block outputs · case 2 GND amp	DF **X5** MAX98357A · case 6 GND twisted with VSYS; the orange band tells it from the right cell B+ / B− pair in the tubes	left tube + middle tube	**48** → 51

6 Middle tube and right middle

screen 2, amplifier, speaker, AM312, P2 · 28 wires finished · housings to fill: X2 F, X4 F, X5 F, X6 F, X7 F, X7 M, X13 M

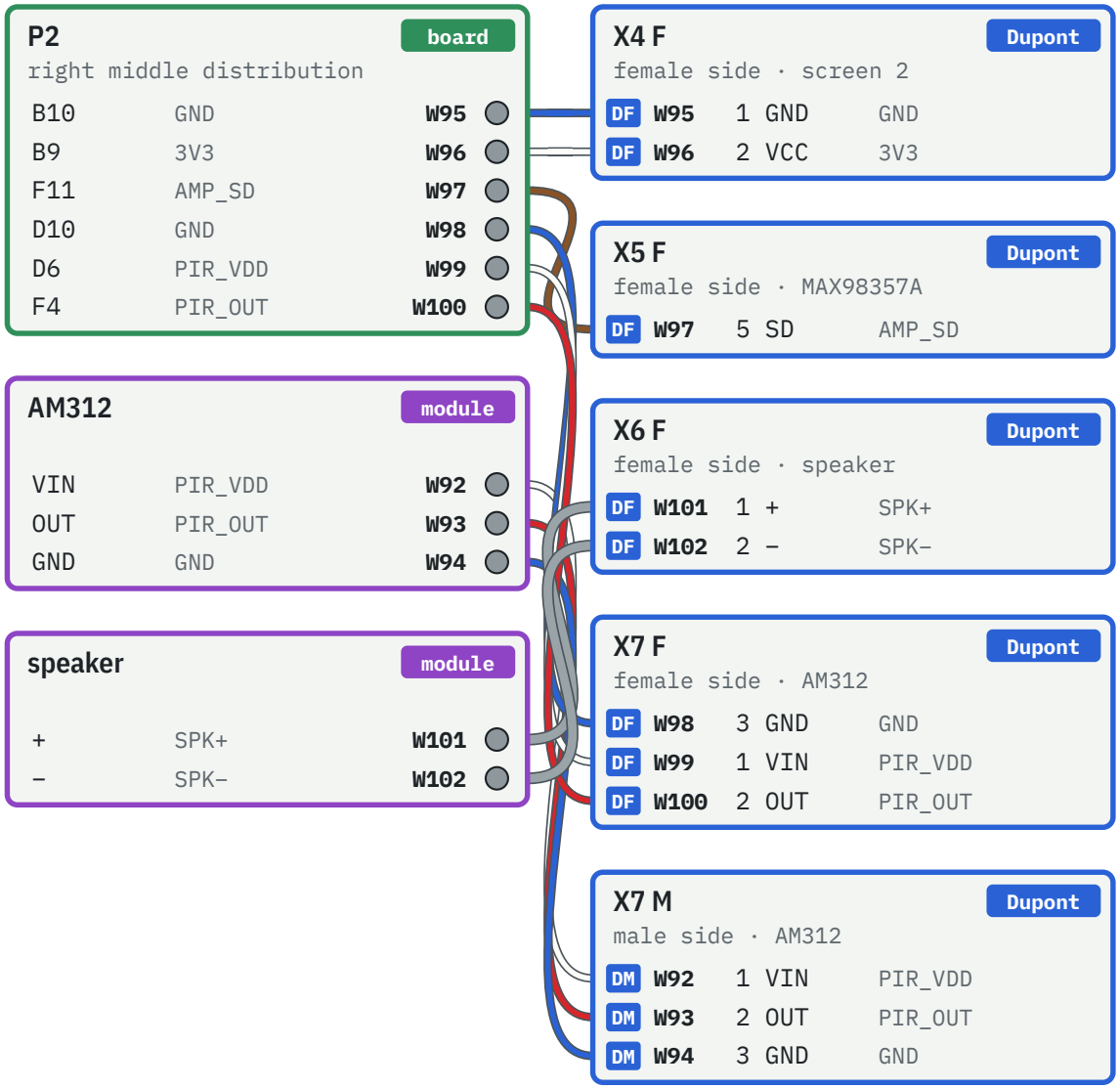
Before

1. Plan renders/tars4_board_p2.svg. Right-angle header on the back of screen 2 (X4). Amplifier: 1 × 4 (GAIN removed) + 1 × 3 headers (X5), 2-pin header instead of the screw terminal (X6), C7 + on VIN, band (–) to GND.
2. AM312: 3-wire pigtail (male contacts, X7), breakout pins trimmed flush, breakout slipped into a heat-shrink sock (it passes 5 mm from the middle tube exit).
3. Thread the middle tube onto the bare wires, then fit it (section Joints) BEFORE soldering their P2 ends and crimping their X4 / X5 ends: this step is tested with the two middle columns joined.

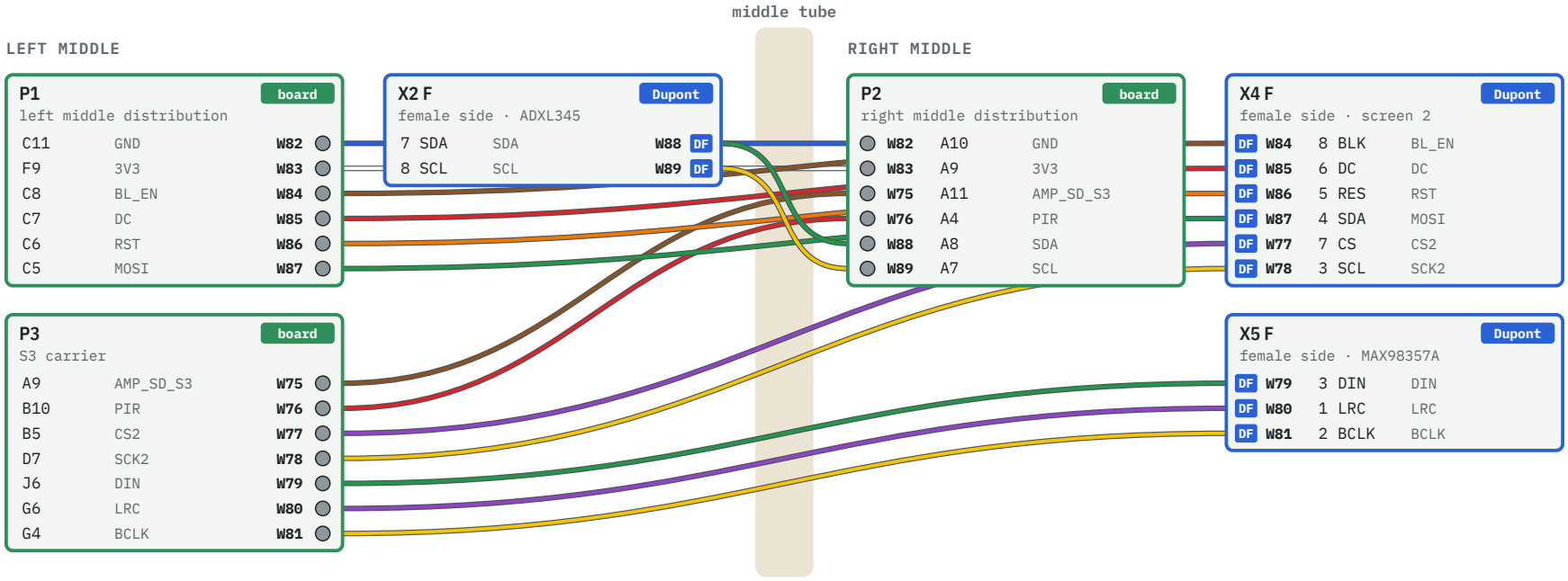
Bench test

- ☐ Before any power-up: every still-free end capped with closed heat-shrink (shrunk, end pinched while hot).
- ☐ Screen 2 displays; sound in the speaker; AMP_SD low cuts the amplifier; PIR rises when a hand passes.

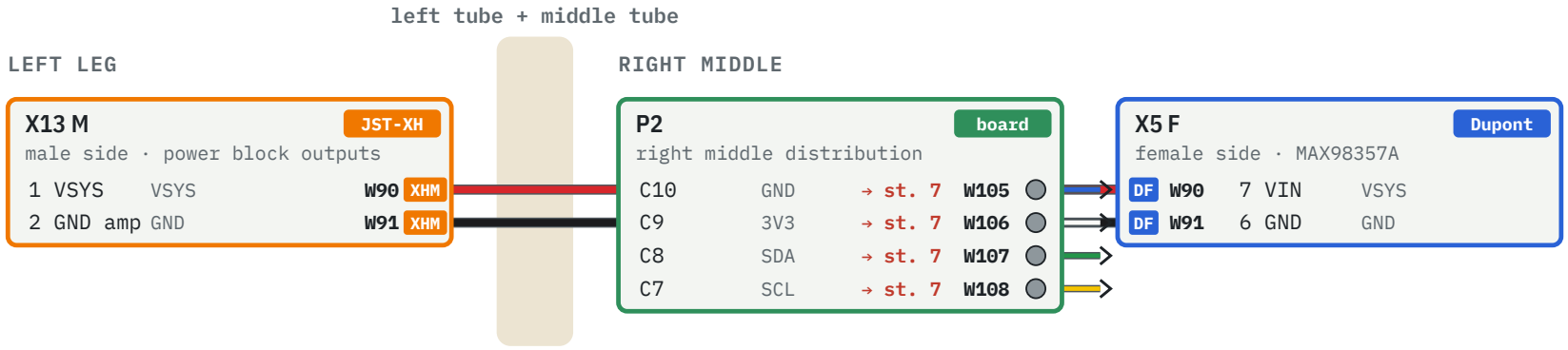
RIGHT MIDDLE



Step 6 · Middle tube and right middle: diagram 2/3: wires through middle tube



Step 6 · Middle tube and right middle: diagram 3/3: wires through right tube, through left tube + middle tube



Wires finished at this step (28)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut
W75	brown 30 AWG, 1 ring	AMP_SD_S3	**P3** S3 carrier · hole A9 (wire 4 of the plan)	**P2** right middle distribution · hole A11 (wire 1 of the plan)	middle tube	**49** → 52
W76	red 30 AWG, 1 ring	PIR	**P3** S3 carrier · hole B10 (wire 9 of the plan)	**P2** right middle distribution · hole A4 (wire 15 of the plan)	middle tube	**51** → 54
W77	purple 30 AWG	CS2	**P3** S3 carrier · hole B5 (wire 14 of the plan)	DF **X4** screen 2 · case 7 CS	middle tube	**47** → 50
W78	yellow 30 AWG	SCK2	**P3** S3 carrier · hole D7 (wire 17 of the plan)	DF **X4** screen 2 · case 3 SCL	middle tube	**46** → 49
W79	green 30 AWG, 1 ring	DIN	**P3** S3 carrier · hole J6 (wire 24 of the plan)	DF **X5** MAX98357A · case 3 DIN	middle tube	**57** → 60
W80	purple 30 AWG, 1 ring	LRC	**P3** S3 carrier · hole G6 (wire 26 of the plan)	DF **X5** MAX98357A · case 1 LRC	middle tube	**58** → 61
W81	yellow 30 AWG, 1 ring	BCLK	**P3** S3 carrier · hole G4 (wire 27 of the plan)	DF **X5** MAX98357A · case 2 BCLK	middle tube	**59** → 62
W82	blue 30 AWG	GND	**P1** left middle distribution · hole C11 (wire 5 of the plan)	**P2** right middle distribution · hole A10 (wire 3 of the plan)	middle tube	**29** → 32
W83	white 30 AWG	3V3	**P1** left middle distribution · hole F9 (wire 11 of the plan)	**P2** right middle distribution · hole A9 (wire 7 of the plan)	middle tube	**29** → 32
W84	brown 30 AWG	BL_EN	**P1** left middle distribution · hole C8 (wire 14 of the plan)	DF **X4** screen 2 · case 8 BLK	middle tube	**27** → 30
W85	red 30 AWG	DC	**P1** left middle distribution · hole C7 (wire 17 of the plan)	DF **X4** screen 2 · case 6 DC	middle tube	**27** → 30
W86	orange 30 AWG	RST	**P1** left middle distribution · hole C6 (wire 20 of the plan)	DF **X4** screen 2 · case 5 RES	middle tube	**27** → 30
W87	green 30 AWG	MOSI	**P1** left middle distribution · hole C5 (wire 23 of the plan)	DF **X4** screen 2 · case 4 SDA	middle tube	**28** → 31
W88	green 30 AWG, 2 rings	SDA	**P2** right middle distribution · hole A8 (wire 10 of the plan)	DF **X2** ADXL345 · case 7 SDA	middle tube	**36** → 39
W89	yellow 30 AWG, 2 rings	SCL	**P2** right middle distribution · hole A7 (wire 12 of the plan)	DF **X2** ADXL345 · case 8 SCL	middle tube	**37** → 40
W90	red 26 AWG, orange band	VSYS	XHM **X13** power block outputs · case 1 VSYS	DF **X5** MAX98357A · case 7 VIN amplifier pair: threaded through the left tube at step 5, X5 end crimped here; twisted with its GND	left tube + middle tube	**48** → 51
W91	black 26 AWG, orange band	GND	XHM **X13** power block outputs · case 2 GND amp	DF **X5** MAX98357A · case 6 GND twisted with VSYS; the orange band tells it from the right cell B+ / B− pair in the tubes	left tube + middle tube	**48** → 51
W92	white 30 AWG	PIR_VDD	AM312 · VIN	DM **X7** AM312 · case 1 VIN 5–6 cm pigtail	—	**6** → 7
W93	red 30 AWG, 1 ring	PIR_OUT	AM312 · OUT	DM **X7** AM312 · case 2 OUT	—	**6** → 7
W94	blue 30 AWG	GND	AM312 · GND	DM **X7** AM312 · case 3 GND	—	**6** → 7
W95	blue 30 AWG	GND	**P2** right middle distribution · hole B10 (wire 4 of the plan)	DF **X4** screen 2 · case 1 GND	—	**14** → 17

W96	white 30 AWG	3V3	**P2** right middle distribution · hole B9 (wire 8 of the plan)	DF **X4** screen 2 · case 2 VCC	—	**15** → **18**
W97	brown 30 AWG, 1 ring	AMP_SD	**P2** right middle distribution · hole F11 (wire 2 of the plan)	DF **X5** MAX98357A · case 5 SD	—	**16** → **19**
W98	blue 30 AWG	GND	**P2** right middle distribution · hole D10 (wire 6 of the plan)	DF **X7** AM312 · case 3 GND	—	**14** → **17**
W99	white 30 AWG	PIR_VDD	**P2** right middle distribution · hole D6 (wire 14 of the plan)	DF **X7** AM312 · case 1 VIN	—	**15** → **18**
W100	red 30 AWG, 1 ring	PIR_OUT	**P2** right middle distribution · hole F4 (wire 16 of the plan)	DF **X7** AM312 · case 2 OUT	—	**17** → **20**
W101	grey 26 AWG	SPK+	speaker · +	DF **X6** speaker · case 1 + + marked with a pen at both ends	—	**11** → **14**
W102	grey 26 AWG	SPK–	speaker · –	DF **X6** speaker · case 2 –	—	**11** → **14**

Wires started here, finished later (4)

One end only now; the other stays bare until the step shown.

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W105	**6→7**	blue 30 AWG	GND	**P2** right middle distribution · hole C10 (wire 5 of the plan)	DF **X9** BME280 · case 2 GND	right tube	**27** → **30**
W106	**6→7**	white 30 AWG	3V3	**P2** right middle distribution · hole C9 (wire 9 of the plan)	DF **X9** BME280 · case 1 VCC	right tube	**27** → **30**
W107	**6→7**	green 30 AWG, 2 rings	SDA	**P2** right middle distribution · hole C8 (wire 11 of the plan)	DF **X9** BME280 · case 4 SDA	right tube	**27** → **30**
W108	**6→7**	yellow 30 AWG, 2 rings	SCL	**P2** right middle distribution · hole C7 (wire 13 of the plan)	DF **X9** BME280 · case 3 SCL	right tube	**28** → **31**

7 Right tube and right leg

DS3231, BME280, right cell · 10 wires finished · housings to fill: X8 F, X9 F, X16 F

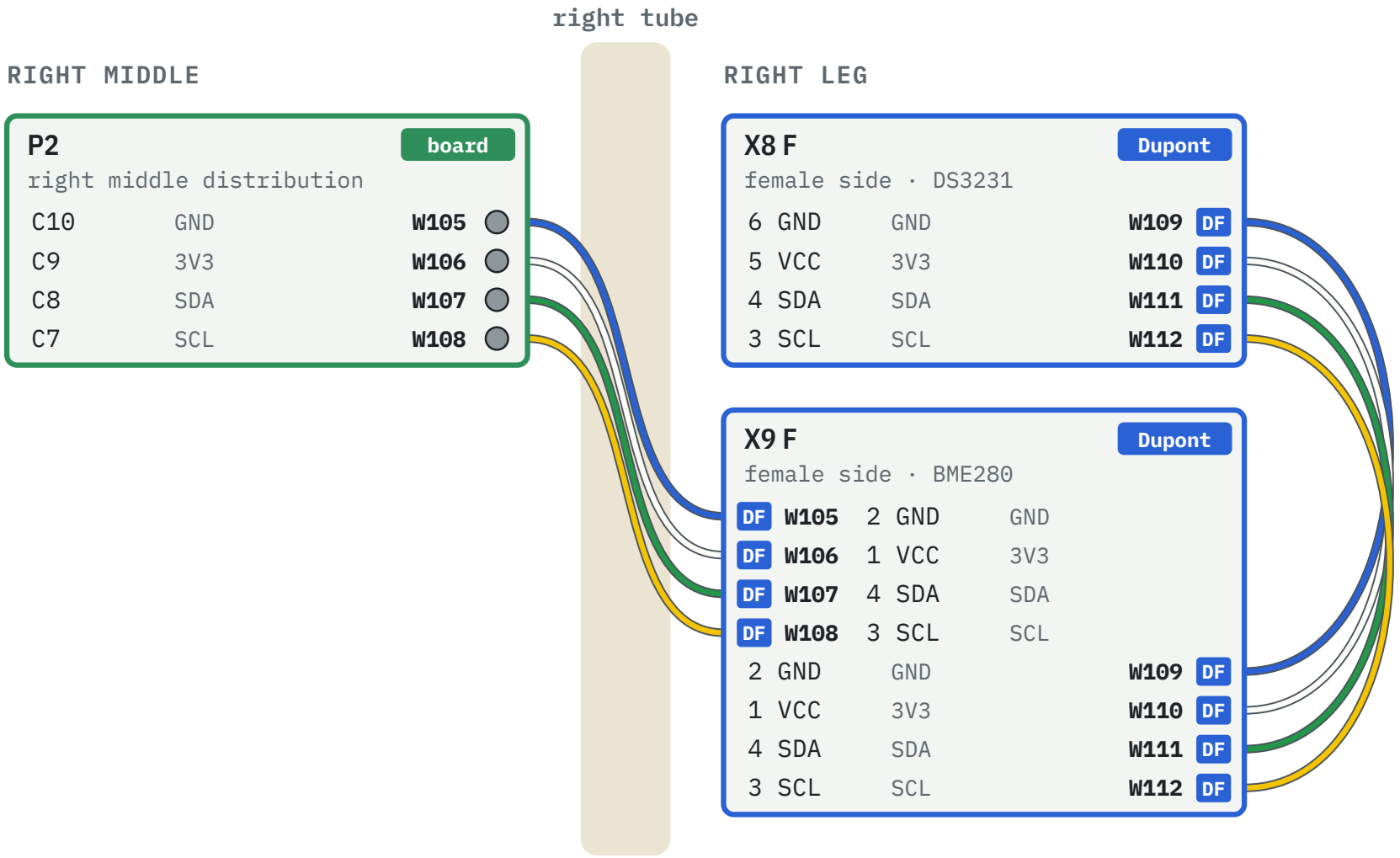
Before

1. DS3231: right-angle header, 32K removed, coin-cell charging resistor removed. BME280: 1 × 4 header, LED removed.
2. Thread the right tube onto the bare wires and fit it (section Joints) BEFORE crimping X9 and soldering the right cell wires: this step is tested with the right leg joined.
3. F2 on the right + plate, then solder the two right cell wires (step 2).
4. X9: each contact takes two wires (P2 + the continuation to X8), crimped together without folding back.
5. Right tube: twist SDA with GND and SCL with 3V3 (long I2C bus, next to SCK / BCLK).

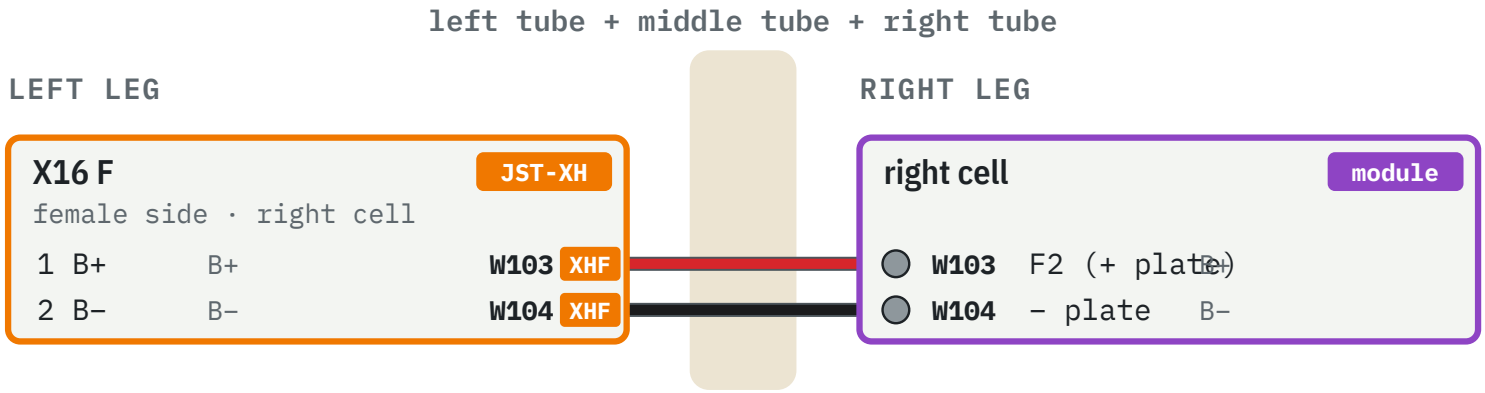
Bench test

- ☐ I2C scan: 0x53 (ADXL345), 0x57 (DS3231 EEPROM), 0x68 (DS3231), 0x76 or 0x77 (BME280).
- ☐ Before plugging in the right cell (X16): same voltage as the left one within 0.05 V.

Step 7 · Right tube and right leg: diagram 1/2: local wires, through right tube



Step 7 · Right tube and right leg: diagram 2/2: wires through left tube + middle tube + right tube



Wires finished at this step (10)

wire	colour	signal	end A	end B	tubes	length cm estimated → to cut
W103	red 26 AWG	B+	right cell · F2 (+ plate)	XHF **X16** right cell · case 1 B+ X16 crimped at step 2, bare end threaded through the 3 tubes, soldered on F2 at step 7; twisted with B−	left tube + middle tube + right tube	**52** → 55
W104	black 26 AWG	B−	right cell · − plate	XHF **X16** right cell · case 2 B− as B+; twisted with B+	left tube + middle tube + right tube	**57** → 60
W105	blue 30 AWG	GND	**P2** right middle distribution · hole C10 (wire 5 of the plan)	DF **X9** BME280 · case 2 GND	right tube	**27** → 30
W106	white 30 AWG	3V3	**P2** right middle distribution · hole C9 (wire 9 of the plan)	DF **X9** BME280 · case 1 VCC	right tube	**27** → 30
W107	green 30 AWG, 2 rings	SDA	**P2** right middle distribution · hole C8 (wire 11 of the plan)	DF **X9** BME280 · case 4 SDA	right tube	**27** → 30
W108	yellow 30 AWG, 2 rings	SCL	**P2** right middle distribution · hole C7 (wire 13 of the plan)	DF **X9** BME280 · case 3 SCL	right tube	**28** → 31
W109	blue 30 AWG	GND	DF **X9** BME280 · case 2 GND	DF **X8** DS3231 · case 6 GND double contact in X9 (with the wire from P2)	—	**14** → 17
W110	white 30 AWG	3V3	DF **X9** BME280 · case 1 VCC	DF **X8** DS3231 · case 5 VCC double contact in X9 (with the wire from P2)	—	**14** → 17
W111	green 30 AWG, 2 rings	SDA	DF **X9** BME280 · case 4 SDA	DF **X8** DS3231 · case 4 SDA double contact in X9 (with the wire from P2)	—	**14** → 17
W112	yellow 30 AWG, 2 rings	SCL	DF **X9** BME280 · case 3 SCL	DF **X8** DS3231 · case 3 SCL double contact in X9 (with the wire from P2)	—	**14** → 17

Joints: fitting the tubes

One brass tube 8 / 7 per joint, cut to 16.5 mm and deburred inside and out, and one PETG collar Ø13 × 2. In each joint the left-hand column is **fixed**: the tube is glued in its teardrop bore Ø8.00, its right face carries 24 detent dimples. The right-hand column **turns** on the tube (bore Ø8.05), on a Ø16 bearing ring with 2 bumps. The collar, glued on the tube inside the turning column, holds the joint together.

A wire with a board on its end does not go through a wall bore: at fitting, one end of every wire in the tube is loose (bare, or a crimped contact out of its housing: they pass the Ø8 bore a few at a time). That side sets the direction the tube is pushed. Fit the tubes left to right.

tube	loose ends	push the tube	collar
left (left leg ↔ left middle)	left leg: the tube-side contacts of X13, X14, X15, X16, out of their housings	from inside the left middle, towards the left leg	slipped on over the loose ends BEFORE the tube goes in: it rides on the tube in the left middle
middle (left middle ↔ right middle)	right middle: P2 ends not soldered yet, X4 / X5 contacts not crimped, right cell pair bare	from inside the left middle, towards the right middle	slipped on over the loose ends once the tube is in place
right (right middle ↔ right leg)	right leg: X9 contacts not crimped, right cell pair bare	from inside the right middle, towards the right leg	slipped on over the loose ends once the tube is in place

For each joint

1. First joint only, a dry run without glue: tube in, collar against the boss, the leg clicks every 15° by hand (the walls flex to let the bumps out). A tight turning bore is reamed by hand with an 8 mm drill.
2. Both columns side by side, the bearing ring against the dimpled face. Thread the loose ends through the two bores, ahead of the tube (a few contacts at a time), then push the tube until its end in the fixed column is flush with the inner face of the boss there.
3. Fixed side: slow epoxy, a fillet around the tube where it leaves the boss inside the fixed column. Never in the gap between the columns nor in the turning bore. Let it cure.
4. Collar: columns pressed together (bumps in their dimples), collar against the turning column’s boss, a drop of epoxy between collar and tube only, never on the boss. The tube sticks out 0.5 mm past it.
5. Swing the leg: a click every step, no side play. Then the contacts back in their housings (connector cards below), the loose ends soldered or crimped, a short heat-shrink sleeve on the bundle at each tube exit.

Wire lengths

Taken from the 3D model (tars4.scad): route from one end to the other along the column, through the pivot (z 238) when the wire changes column, +10 % for bends, + service loop at each end (7 cm at the P3 carrier, 4 cm at connectors, 2–3 cm at boards and modules), + 3 cm per tube (leg rotation). Length to cut = estimated + 3 cm; module pigtails are 6 cm. Cut at assembly time, after offering the wire up in TARS if possible.

colour	gauge	wires	to cut
brown	30 AWG	12	3.67 m
yellow	30 AWG	11	3.61 m
green	30 AWG	12	3.49 m
purple	30 AWG	7	2.49 m
blue	30 AWG	12	2.24 m
white	30 AWG	13	2.21 m
red	30 AWG	7	1.83 m
orange	30 AWG	5	1.45 m
red	26 AWG	15	3.32 m
black	26 AWG	10	2.49 m
blue	26 AWG	3	0.83 m
white	26 AWG	3	0.81 m
grey	26 AWG	2	0.28 m
total 30 AWG			20.99 m
total 26 AWG			7.73 m

Full wire list

step where it is started → finished

wire	step	colour	signal	end A	end B	tubes	length cm estimated → to cut
W1	1	 red 26 AWG, purple band	VUSB	● **P4** power path · hole F8 (wire 1 of the plan)	● TP4056 · IN+ (VUSB)	–	**10** → **13**
W2	1	 red 26 AWG, white band	OUT+	● **P4** power path · hole B4 (wire 2 of the plan)	● TP4056 · OUT+	–	**11** → **14**
W3	1	 black 26 AWG, white band	GND	● **P4** power path · hole C1 (wire 7 of the plan)	● TP4056 · OUT–	–	**11** → **14**
W4	1	 brown 30 AWG	VBAT/2 → B+	● **P4** power path · hole J5 (wire 10 of the plan)	● TP4056 · B+ pad (R2 1 MΩ in line at the pad end) R2 1 MΩ soldered in line ~2 cm from the other end, in heat-shrink	–	**11** → **14**
W5	1	 red 26 AWG, orange band	VSYS	● **P4** power path · hole C3 (wire 5 of the plan)	● TPS63020 · VIN	–	**12** → **15**
W6	1	 black 26 AWG, orange band	GND	● **P4** power path · hole D1 (wire 8 of the plan)	● TPS63020 · GND	–	**12** → **15**
W7	1	 white 30 AWG	3V3	● **P4** power path · hole H8 (wire 13 of the plan)	● TPS63020 · VOUT	–	**12** → **15**
W8	1	 red 26 AWG, yellow band	VPATH	● **P4** power path · hole A8 (wire 3 of the plan)	XHF **X10** J1 + SW1 · case 3 VPATH	–	**11** → **14**
W9	1	 red 26 AWG, orange band	VSYS	● **P4** power path · hole D3 (wire 4 of the plan)	XHF **X10** J1 + SW1 · case 4 VSYS	–	**10** → **13**
W10	1	 red 26 AWG, orange band	VSYS	● **P4** power path · hole C3 (wire 6 of the plan)	XHF **X13** power block outputs · case 1 VSYS	–	**11** → **14**
W11	1	 black 26 AWG, orange band	GND	● **P4** power path · hole E1 (wire 9 of the plan)	XHF **X13** power block outputs · case 2 GND amp	–	**11** → **14**
W12	1	 brown 30 AWG	VBAT/2	● **P4** power path · hole J3 (wire 11 of the plan)	XHF **X14** measurements · case 1 VBAT/2	–	**11** → **14**
W13	1	 orange 30 AWG	USB_DET	● **P4** power path · hole E2 (wire 12 of the plan)	XHF **X14** measurements · case 2 USB_DET	–	**9** → **12**
W14	1	 brown 30 AWG, 3 rings	SD_EN	● **P4** power path · hole H5 (wire 15 of the plan)	XHF **X14** measurements · case 3 SD_EN	–	**11** → **14**
W15	1	 brown 30 AWG, 1 ring	VSYS → VSENSE_B	● **P4** power path · hole D4 (wire 16 of the plan)	XHF **X14** measurements · case 4 VSENSE_B R23 1 MΩ soldered in line ~2 cm from the board end, in heat-shrink	–	**10** → **13**
W16	1	 red 26 AWG, purple band	VUSB	● TP4056 · IN+	XHF **X10** J1 + SW1 · case 1 VUSB	–	**9** → **12**
W17	1	 black 26 AWG, purple band	GND	● TP4056 · IN–	XHF **X10** J1 + SW1 · case 2 GND	–	**9** → **12**
W18	1	 red 26 AWG	B+	● TP4056 · B+	XHM **X12** left cell · case 1 B+ cell junction: two wires soldered on the B+ pad	–	**10** → **13**

W19	**1**	black 26 AWG	B−	TP4056 · B−	XHM **X12** left cell · case 2 B− cell junction: two wires soldered on the B− pad	—	**10** → **13**
W20	**1**	red 26 AWG	B+	TP4056 · B+	XHM **X16** right cell · case 1 B+ to X16, parked above the power-path board	—	**13** → **16**
W21	**1**	black 26 AWG	B−	TP4056 · B−	XHM **X16** right cell · case 2 B−	—	**13** → **16**
W22	**1**	white 26 AWG	3V3	TPS63020 · VOUT	XHF **X13** power block outputs · case 3 3V3 along the logic GND wire	—	**11** → **14**
W23	**1**	blue 26 AWG	GND	TPS63020 · GND	XHF **X13** power block outputs · case 4 GND log logic GND, along the 3V3	—	**11** → **14**
W24	**2**	red 26 AWG, purple band	VUSB	J1 micro-USB · VBUS	XHM **X10** J1 + SW1 · case 1 VUSB twisted with the GND	—	**24** → **27**
W25	**2**	black 26 AWG, purple band	GND	J1 micro-USB · GND	XHM **X10** J1 + SW1 · case 2 GND	—	**24** → **27**
W26	**2**	red 26 AWG, yellow band	VPATH	SW1 switch · end lug	XHM **X10** J1 + SW1 · case 3 VPATH	—	**24** → **27**
W27	**2**	red 26 AWG, orange band	VSYS	SW1 switch · centre lug	XHM **X10** J1 + SW1 · case 4 VSYS	—	**24** → **27**
W28	**2**	red 26 AWG	B+	left cell · F1 (+ plate)	XHF **X12** left cell · case 1 B+ twisted with B−	—	**18** → **21**
W29	**2**	black 26 AWG	B−	left cell · − plate	XHF **X12** left cell · case 2 B−	—	**24** → **27**
W30	**3→4**	white 26 AWG	3V3	**P3** S3 carrier · hole A4 (wire 1 of the plan)	**P1** left middle distribution · hole B9 (wire 8 of the plan)	—	**28** → **31**
W31	**3→4**	blue 26 AWG	GND	**P3** S3 carrier · hole J5 (wire 2 of the plan)	**P1** left middle distribution · hole B10 (wire 2 of the plan)	—	**30** → **33**
W32	**3→4**	red 30 AWG	DC	**P3** S3 carrier · hole A7 (wire 5 of the plan)	**P1** left middle distribution · hole A7 (wire 15 of the plan)	—	**26** → **29**
W33	**3→4**	green 30 AWG	MOSI	**P3** S3 carrier · hole C10 (wire 6 of the plan)	**P1** left middle distribution · hole A5 (wire 21 of the plan)	—	**25** → **28**
W34	**3→4**	brown 30 AWG	BL_EN	**P3** S3 carrier · hole B7 (wire 12 of the plan)	**P1** left middle distribution · hole A8 (wire 12 of the plan)	—	**27** → **30**
W35	**3→4**	orange 30 AWG	RST	**P3** S3 carrier · hole B6 (wire 13 of the plan)	**P1** left middle distribution · hole A6 (wire 18 of the plan)	—	**26** → **29**
W36	**3→4**	purple 30 AWG	CS1	**P3** S3 carrier · hole A5 (wire 7 of the plan)	DF **X1** screen 1 · case 7 CS	—	**36** → **39**
W37	**3→4**	yellow 30 AWG	SCK1	**P3** S3 carrier · hole C7 (wire 16 of the plan)	DF **X1** screen 1 · case 3 SCL	—	**35** → **38**
W38	**3→4**	red 30 AWG	INT1	**P3** S3 carrier · hole A10 (wire 3 of the plan)	DF **X2** ADXL345 · case 4 INT1	—	**22** → **25**
W39	**3→4**	green 30 AWG, 2 rings	SDA	**P3** S3 carrier · hole B9 (wire 10 of the plan)	DF **X2** ADXL345 · case 7 SDA	—	**22** → **25**
W40	**3→4**	yellow 30 AWG, 2 rings	SCL	**P3** S3 carrier · hole B8 (wire 11 of the plan)	DF **X2** ADXL345 · case 8 SCL	—	**22** → **25**
W41	**3→4**	yellow 30 AWG, 1 ring	MIC_BCLK	**P3** S3 carrier · hole B4 (wire 15 of the plan)	DF **X3** INMP441 · case 1 SCK	—	**24** → **27**

W42	**3→4**	purple 30 AWG, 1 ring	MIC_WS	**P3** S3 carrier · hole I9 (wire 18 of the plan)	DF **X3** INMP441 · case 2 WS	–	**22** → **25**
W43	**3→4**	green 30 AWG, 1 ring	MIC_SD	**P3** S3 carrier · hole I4 (wire 21 of the plan)	DF **X3** INMP441 · case 3 SD	–	**23** → **26**
W44	**4**	yellow 30 AWG, 1 ring	MIC_BCLK	INMP441 · SCK	DM **X3** INMP441 · case 1 SCK 5–6 cm pigtail	–	**6** → **7**
W45	**4**	purple 30 AWG, 1 ring	MIC_WS	INMP441 · WS	DM **X3** INMP441 · case 2 WS	–	**6** → **7**
W46	**4**	green 30 AWG, 1 ring	MIC_SD	INMP441 · SD	DM **X3** INMP441 · case 3 SD	–	**6** → **7**
W47	**4**	white 30 AWG	3V3	INMP441 · VDD	DM **X3** INMP441 · case 4 VDD	–	**6** → **7**
W48	**4**	blue 30 AWG	GND	INMP441 · GND	DM **X3** INMP441 · case 5 GND L/R bridged to GND on the board	–	**6** → **7**
W49	**4**	blue 30 AWG	GND	**P1** left middle distribution · hole A10 (wire 1 of the plan)	DF **X1** screen 1 · case 1 GND	–	**15** → **18**
W50	**4**	white 30 AWG	3V3	**P1** left middle distribution · hole A9 (wire 7 of the plan)	DF **X1** screen 1 · case 2 VCC	–	**15** → **18**
W51	**4**	brown 30 AWG	BL_EN	**P1** left middle distribution · hole B8 (wire 13 of the plan)	DF **X1** screen 1 · case 8 BLK	–	**15** → **18**
W52	**4**	red 30 AWG	DC	**P1** left middle distribution · hole B7 (wire 16 of the plan)	DF **X1** screen 1 · case 6 DC	–	**15** → **18**
W53	**4**	orange 30 AWG	RST	**P1** left middle distribution · hole B6 (wire 19 of the plan)	DF **X1** screen 1 · case 5 RES	–	**15** → **18**
W54	**4**	green 30 AWG	MOSI	**P1** left middle distribution · hole B5 (wire 22 of the plan)	DF **X1** screen 1 · case 4 SDA	–	**16** → **19**
W55	**4**	blue 30 AWG	GND	**P1** left middle distribution · hole D11 (wire 6 of the plan)	DF **X3** INMP441 · case 5 GND	–	**16** → **19**
W56	**4**	white 30 AWG	3V3	**P1** left middle distribution · hole D9 (wire 10 of the plan)	DF **X3** INMP441 · case 4 VDD	–	**15** → **18**
W57	**4**	white 30 AWG	3V3	DF **X3** INMP441 · case 4 VDD	DF **X2** ADXL345 · case 2 VCC double contact in X3 (with the wire from P1)	–	**12** → **15**
W58	**4**	blue 30 AWG	GND	DF **X3** INMP441 · case 5 GND	DF **X2** ADXL345 · case 1 GND double contact in X3 (with the wire from P1)	–	**12** → **15**
W59	**1→5**	white 30 AWG, 3 rings	SD_3V3	**P4** power path · hole I6 (wire 14 of the plan)	DF **X15** microSD · case 4 3V3	–	**16** → **19**
W60	**3→5**	brown 30 AWG	VBAT/2	**P3** S3 carrier · hole B11 (wire 8 of the plan)	XHM **X14** measurements · case 1 VBAT/2	left tube	**50** → **53**
W61	**3→5**	orange 30 AWG	USB_DET	**P3** S3 carrier · hole I7 (wire 20 of the plan)	XHM **X14** measurements · case 2 USB_DET	left tube	**53** → **56**
W62	**3→5**	brown 30 AWG, 3 rings	SD_EN	**P3** S3 carrier · hole J9 (wire 22 of the plan)	XHM **X14** measurements · case 3 SD_EN	left tube	**52** → **55**

W63	**3→5**	brown 30 AWG, 1 ring	VSENSE_B	**P3** S3 carrier · hole G8 (wire 28 of the plan)	XHM **X14** measurements · case 4 VSENSE_B	left tube	**52** → 55
W64	**3→5**	purple 30 AWG, 3 rings	SD_CMD	**P3** S3 carrier · hole I8 (wire 19 of the plan)	DF **X15** microSD · case 2 CMD	left tube	**57** → 60
W65	**3→5**	green 30 AWG, 3 rings	SD_D0	**P3** S3 carrier · hole J8 (wire 23 of the plan)	DF **X15** microSD · case 3 D0	left tube	**57** → 60
W66	**3→5**	yellow 30 AWG, 3 rings	SD_CLK	**P3** S3 carrier · hole G11 (wire 25 of the plan)	DF **X15** microSD · case 1 CLK	left tube	**55** → 58
W67	**4→5**	blue 26 AWG	GND	**P1** left middle distribution · hole C10 (wire 3 of the plan)	XHM **X13** power block outputs · case 4 GND log	left tube	**33** → 36
W68	**4→5**	white 26 AWG	3V3	**P1** left middle distribution · hole C9 (wire 9 of the plan)	XHM **X13** power block outputs · case 3 3V3	left tube	**33** → 36
W69	**4→5**	blue 30 AWG, 3 rings	GND	**P1** left middle distribution · hole B11 (wire 4 of the plan)	DF **X15** microSD · case 5 GND	left tube	**35** → 38
W70	**5**	yellow 30 AWG, 3 rings	SD_CLK	microSD reader · CLK	DM **X15** microSD · case 1 CLK 5–6 cm pigtail	—	**6** → 7
W71	**5**	purple 30 AWG, 3 rings	SD_CMD	microSD reader · CMD	DM **X15** microSD · case 2 CMD	—	**6** → 7
W72	**5**	green 30 AWG, 3 rings	SD_D0	microSD reader · D0	DM **X15** microSD · case 3 D0	—	**6** → 7
W73	**5**	white 30 AWG, 3 rings	SD_3V3	microSD reader · VCC	DM **X15** microSD · case 4 3V3	—	**6** → 7
W74	**5**	blue 30 AWG, 3 rings	GND	microSD reader · GND	DM **X15** microSD · case 5 GND	—	**6** → 7
W75	**3→6**	brown 30 AWG, 1 ring	AMP_SD_S3	**P3** S3 carrier · hole A9 (wire 4 of the plan)	**P2** right middle distribution · hole A11 (wire 1 of the plan)	middle tube	**49** → 52
W76	**3→6**	red 30 AWG, 1 ring	PIR	**P3** S3 carrier · hole B10 (wire 9 of the plan)	**P2** right middle distribution · hole A4 (wire 15 of the plan)	middle tube	**51** → 54
W77	**3→6**	purple 30 AWG	CS2	**P3** S3 carrier · hole B5 (wire 14 of the plan)	DF **X4** screen 2 · case 7 CS	middle tube	**47** → 50
W78	**3→6**	yellow 30 AWG	SCK2	**P3** S3 carrier · hole D7 (wire 17 of the plan)	DF **X4** screen 2 · case 3 SCL	middle tube	**46** → 49
W79	**3→6**	green 30 AWG, 1 ring	DIN	**P3** S3 carrier · hole J6 (wire 24 of the plan)	DF **X5** MAX98357A · case 3 DIN	middle tube	**57** → 60
W80	**3→6**	purple 30 AWG, 1 ring	LRC	**P3** S3 carrier · hole G6 (wire 26 of the plan)	DF **X5** MAX98357A · case 1 LRC	middle tube	**58** → 61
W81	**3→6**	yellow 30 AWG, 1 ring	BCLK	**P3** S3 carrier · hole G4 (wire 27 of the plan)	DF **X5** MAX98357A · case 2 BCLK	middle tube	**59** → 62
W82	**4→6**	blue 30 AWG	GND	**P1** left middle distribution · hole C11 (wire 5 of the plan)	**P2** right middle distribution · hole A10 (wire 3 of the plan)	middle tube	**29** → 32
W83	**4→6**	white 30 AWG	3V3	**P1** left middle distribution · hole F9 (wire 11 of the plan)	**P2** right middle distribution · hole A9 (wire 7 of the plan)	middle tube	**29** → 32

W84	4→6	brown 30 AWG	BL_EN	**P1** left middle distribution · hole C8 (wire 14 of the plan)	DF **X4** screen 2 · case 8 BLK	middle tube	**27** → **30**
W85	4→6	red 30 AWG	DC	**P1** left middle distribution · hole C7 (wire 17 of the plan)	DF **X4** screen 2 · case 6 DC	middle tube	**27** → **30**
W86	4→6	orange 30 AWG	RST	**P1** left middle distribution · hole C6 (wire 20 of the plan)	DF **X4** screen 2 · case 5 RES	middle tube	**27** → **30**
W87	4→6	green 30 AWG	MOSI	**P1** left middle distribution · hole C5 (wire 23 of the plan)	DF **X4** screen 2 · case 4 SDA	middle tube	**28** → **31**
W88	4→6	green 30 AWG, 2 rings	SDA	**P2** right middle distribution · hole A8 (wire 10 of the plan)	DF **X2** ADXL345 · case 7 SDA	middle tube	**36** → **39**
W89	4→6	yellow 30 AWG, 2 rings	SCL	**P2** right middle distribution · hole A7 (wire 12 of the plan)	DF **X2** ADXL345 · case 8 SCL	middle tube	**37** → **40**
W90	5→6	red 26 AWG, orange band	VSYS	XHM **X13** power block outputs · case 1 VSYS	DF **X5** MAX98357A · case 7 VIN amplifier pair: threaded through the left tube at step 5, X5 end crimped here; twisted with its GND	left tube + middle tube	**48** → **51**
W91	5→6	black 26 AWG, orange band	GND	XHM **X13** power block outputs · case 2 GND amp	DF **X5** MAX98357A · case 6 GND twisted with VSYS; the orange band tells it from the right cell B+ / B− pair in the tubes	left tube + middle tube	**48** → **51**
W92	6	white 30 AWG	PIR_VDD	AM312 · VIN	DM **X7** AM312 · case 1 VIN 5–6 cm pigtail	—	**6** → **7**
W93	6	red 30 AWG, 1 ring	PIR_OUT	AM312 · OUT	DM **X7** AM312 · case 2 OUT	—	**6** → **7**
W94	6	blue 30 AWG	GND	AM312 · GND	DM **X7** AM312 · case 3 GND	—	**6** → **7**
W95	6	blue 30 AWG	GND	**P2** right middle distribution · hole B10 (wire 4 of the plan)	DF **X4** screen 2 · case 1 GND	—	**14** → **17**
W96	6	white 30 AWG	3V3	**P2** right middle distribution · hole B9 (wire 8 of the plan)	DF **X4** screen 2 · case 2 VCC	—	**15** → **18**
W97	6	brown 30 AWG, 1 ring	AMP_SD	**P2** right middle distribution · hole F11 (wire 2 of the plan)	DF **X5** MAX98357A · case 5 SD	—	**16** → **19**
W98	6	blue 30 AWG	GND	**P2** right middle distribution · hole D10 (wire 6 of the plan)	DF **X7** AM312 · case 3 GND	—	**14** → **17**
W99	6	white 30 AWG	PIR_VDD	**P2** right middle distribution · hole D6 (wire 14 of the plan)	DF **X7** AM312 · case 1 VIN	—	**15** → **18**
W100	6	red 30 AWG, 1 ring	PIR_OUT	**P2** right middle distribution · hole F4 (wire 16 of the plan)	DF **X7** AM312 · case 2 OUT	—	**17** → **20**
W101	6	grey 26 AWG	SPK+	speaker · +	DF **X6** speaker · case 1 + + marked with a pen at both ends	—	**11** → **14**
W102	6	grey 26 AWG	SPK−	speaker · −	DF **X6** speaker · case 2 −	—	**11** → **14**
W103	2→7	red 26 AWG	B+	right cell · F2 (+ plate)	XHF **X16** right cell · case 1 B+ X16 crimped at step 2, bare end threaded through the 3 tubes, soldered on F2 at step 7; twisted with B−	left tube + middle tube + right tube	**52** → **55**
W104	2→7	black 26 AWG	B−	right cell · − plate	XHF **X16** right cell · case 2 B− as B+; twisted with B+	left tube + middle tube + right tube	**57** → **60**

W105	**6→7**	blue 30 AWG	GND	● **P2** right middle distribution · hole C10 (wire 5 of the plan)	**DF** **X9** BME280 · case 2 GND	right tube	**27** → **30**
W106	**6→7**	white 30 AWG	3V3	● **P2** right middle distribution · hole C9 (wire 9 of the plan)	**DF** **X9** BME280 · case 1 VCC	right tube	**27** → **30**
W107	**6→7**	green 30 AWG, 2 rings	SDA	● **P2** right middle distribution · hole C8 (wire 11 of the plan)	**DF** **X9** BME280 · case 4 SDA	right tube	**27** → **30**
W108	**6→7**	yellow 30 AWG, 2 rings	SCL	● **P2** right middle distribution · hole C7 (wire 13 of the plan)	**DF** **X9** BME280 · case 3 SCL	right tube	**28** → **31**
W109	**7**	blue 30 AWG	GND	**DF** **X9** BME280 · case 2 GND	**DF** **X8** DS3231 · case 6 GND double contact in X9 (with the wire from P2)	—	**14** → **17**
W110	**7**	white 30 AWG	3V3	**DF** **X9** BME280 · case 1 VCC	**DF** **X8** DS3231 · case 5 VCC double contact in X9 (with the wire from P2)	—	**14** → **17**
W111	**7**	green 30 AWG, 2 rings	SDA	**DF** **X9** BME280 · case 4 SDA	**DF** **X8** DS3231 · case 4 SDA double contact in X9 (with the wire from P2)	—	**14** → **17**
W112	**7**	yellow 30 AWG, 2 rings	SCL	**DF** **X9** BME280 · case 3 SCL	**DF** **X8** DS3231 · case 3 SCL double contact in X9 (with the wire from P2)	—	**14** → **17**

CONNECTORS (HARNESS SIDE, CAVITY 1 ON THE LEFT)

X1 screen 1 · Dupont 1 × 8 on a right-angle header

GND	VCC	SCL	SDA	RES
GND	3V3	SCK1	MOSI	RST

DC	CS	BLK
DC	CS1	BL_EN

X2 ADXL345 · Dupont 1 × 8, 5 wires

GND	VCC	CS	INT1	INT2
GND	3V3	empty	INT1	plugged

SDO	SDA	SCL
empty	SDA	SCL

X3 INMP441 · pigtail + inline Dupont 1 × 6

SCK	WS	SD	VDD	GND
MIC_BCLK	MIC_WS	MIC_SD	3V3	GND

–
plugged

X4 screen 2 · Dupont 1 × 8 on a right-angle header

GND	VCC	SCL	SDA	RES
GND	3V3	SCK2	MOSI	RST

DC	CS	BLK
DC	CS2	BL_EN

X5 MAX98357A · Dupont 1 × 4 + 1 × 3

LRC	BCLK	DIN	GAIN	SD
LRC	BCLK	DIN	plugged	AMP_SD

GND	VIN
GND	VSYS

X6 speaker · Dupont 1 × 2 on a header

+	–
SPK+	SPK–

X7 AM312 · pigtail + inline Dupont 1 × 4

VIN	OUT	GND	–
PIR_VDD	PIR_OUT	GND	plugged

X8 DS3231 · Dupont 1 × 6 on a right-angle header, 4 wires

32K	SQW	SCL	SDA	VCC
plugged	empty	SCL	SDA	3V3

GND
GND

X9 BME280 · Dupont 1 × 4 (daisy-chained to X8)

VCC	GND	SCL	SDA
3V3	GND	SCL	SDA

X10 J1 + SW1 · inline JST-XH 4

VUSB	GND	VPATH	VSYS
VUSB	GND	VPATH	VSYS

X12 left cell · inline JST-XH 2

B+	B–
B+	B–

X13 power block outputs · inline JST-XH 4

VSYS	GND amp	3V3	GND log
VSYS	GND	3V3	GND

X14 measurements · inline JST-XH 4

VBAT/2	USB_DET	SD_EN	VSENSE_B
VBAT/2	USB_DET	SD_EN	VSENSE_B

X15 microSD · pigtail + inline Dupont 1 × 6

CLK	CMD	D0	3V3	GND
SD_CLK	SD_CMD	SD_D0	SD_3V3	GND

–
plugged

X16 right cell · inline JST-XH 2

B+	B–
B+	B–